Practical 10

Consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 41, 122, 14, 124, 65, 67. The SSTF scheduling algorithm is used. The head is initially at cylinder number 53. The cylinders are numbered from 0 to 199. Write a Program to calculate the total head movement (in number of cylinders) incurred while servicing these requests.

```
  GNU nano 8.7
#include <stdio.h>
#include <stdlib.h>

int main() {
    int req[] = {98, 183, 41, 122, 14, 124, 65, 67};
    int n = 8;
    int head = 53;
    int visited[8] = {0};
    int total = 0, i, j;

    printf("Order of servicing:\n");

    for(i = 0; i < n; i++) {
        int min = 9999;
        int index = -1;

        for(j = 0; j < n; j++) {
            if(!visited[j]) {
                int dist = abs(head - req[j]);
                if(dist < min) {
                    min = dist;
                    index = j;
                }
            }
        }

        visited[index] = 1;
        total += min;
        head = req[index];

        printf("%d ", head);
    }

    printf("\nTotal Head Movement = %d\n", total);
                                                          [ Read
```

```
Dell@aryaaa MSYS ~
$ nano disk.c

Dell@aryaaa MSYS ~
$ gcc disk.c -o disk

Dell@aryaaa MSYS ~
$ ./disk
Order of servicing:
41 65 67 98 122 124 183 14
Total Head Movement = 323

Dell@aryaaa MSYS ~
$
```